

Llama 3.2 Model Card — Meta AI

Sumber: https://github.com/meta-llama/llama-models/blob/main/models/llama3_2/MODEL_CARD.md

Tanggal akses: 12 Mei 2026

Llama 3.2 Model Card — Model Information

The Llama 3.2 collection of multilingual large language models (LLMs) is a collection of pretrained and instruction-tuned generative models in 1B and 3B sizes (text in/text out). The Llama 3.2 instruction-tuned text only models are optimized for multilingual dialogue use cases, including agentic retrieval and summarization tasks. They outperform many of the available open source and closed chat models on common industry benchmarks.

Model Developer: Meta

Model Architecture: Llama 3.2 is an auto-regressive language model that uses an optimized transformer architecture. The tuned versions use supervised fine-tuning (SFT) and reinforcement learning with human feedback (RLHF) to align with human preferences for helpfulness and safety.

Model Release Date: October 24, 2024

License: Llama 3.2 Community License (custom commercial license)

Model Variants

Model	Parameters	Input Modalities	Output Modalities	Context Length	Training Data
Llama 3.2 1B (text)	1.23B	Multilingual Text	Multilingual Text & Code	128K	Up to 9T tokens
Llama 3.2 3B (text)	3.21B	Multilingual Text	Multilingual Text & Code	128K	Up to 9T tokens
Llama 3.2 1B Quantized	1.23B	Multilingual Text	Multilingual Text & Code	8K	Up to 9T tokens
Llama 3.2 3B Quantized	3.21B	Multilingual Text	Multilingual Text & Code	8K	Up to 9T tokens

Supported Languages: English, German, French, Italian, Portuguese, Hindi, Spanish, and Thai are officially supported. Llama 3.2 has been trained on a broader collection of languages beyond these 8 supported languages.

Knowledge Cutoff: December 2023

Intended Use

Intended Use Cases: Llama 3.2 is intended for commercial and research use in multiple languages. Instruction tuned text only models are intended for assistant-like chat and agentic applications like knowledge retrieval and summarization, mobile AI powered writing assistants and query and prompt rewriting.

Out of Scope: Use in any manner that violates applicable laws or regulations. Use in any other way that is prohibited by the Acceptable Use Policy and Llama 3.2 Community License.

Training Data

Llama 3.2 was pretrained on up to 9 trillion tokens of data from publicly available sources. For the 1B and 3B models, logits from the Llama 3.1 8B and 70B models were incorporated into the pretraining stage through knowledge distillation. In post-training, each round involved Supervised Fine-Tuning (SFT), Rejection Sampling (RS), and Direct Preference Optimization (DPO).

Hardware and Training

Model	Training Time (GPU hours)	Training Emissions (tons CO ₂ eq)
Llama 3.2 1B	370K	107
Llama 3.2 3B	460K	133
Llama 3.2 1B QLoRA	1.3K	0.381
Llama 3.2 3B QLoRA	1.6K	0.461
Total	833K	240

Training utilized H100-80GB GPUs (TDP 700W). Since 2020, Meta has maintained net zero greenhouse gas emissions in its global operations and matched 100% of its electricity use with renewable energy.

Quantization

Linear layers in all transformer blocks are quantized to 4-bit groupwise (group size 32) for weights and 8-bit per-token dynamic quantization for activations. Two quantization approaches were used: **SpinQuant** (rotation matrix optimization) and **QAT + LoRA** (quantization-aware training with low-rank adaptation).

Benchmark Performance (Instruction Tuned Models)

Benchmark	Metric	Llama 3.2 1B	Llama 3.2 3B	Llama 3.1 8B
MMLU	acc	49.3	63.4	69.4
GSM8K (CoT)	em	44.4	77.7	84.5
MATH (CoT)	em	30.6	48.0	51.9
ARC-C	acc	59.4	78.6	83.4
IFEval	avg acc	59.5	77.4	80.4
HellaSwag	acc	41.2	69.8	78.7

Safety and Responsibility

Meta implemented safety mitigations including supervised fine-tuning, red teaming exercises, and safety evaluations across critical risk areas (CBRNE, Child Safety, Cyber Attacks). Developers are expected to deploy system safeguards (Llama Guard, Prompt Guard, Code Shield) when building agentic systems with Llama 3.2.

Ethical Considerations

Llama 3.2's core values are openness, inclusivity and helpfulness. As with all LLMs, potential outputs cannot be predicted in advance, and the model may produce inaccurate, biased or objectionable responses. Developers should perform safety testing tailored to their specific applications.

Dokumen ini dicetak dari sumber online untuk keperluan referensi tesis.

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